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Role: Computer Vision

AI Workflow Documentation Report

1. Introduction

The AI workflow defines the complete operational architecture of the virtual try-on system. It describes how input images are processed through multiple AI modules to generate realistic try-on outputs.

The workflow includes image preprocessing, pose detection, human parsing, garment warping, rendering, and final output generation. Proper workflow design improves scalability, maintainability, and system performance.

The uploaded workflow notes define the complete pipeline from user upload to rendering.

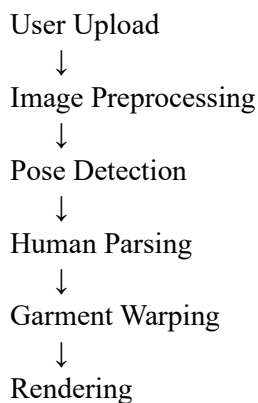
2. Objectives

The objectives of workflow documentation are:

- Define current AI pipeline structure
- Document input and output flow
- Describe individual AI modules
- Analyze dependencies between components
- Identify system challenges
- Propose improvements

3. Overall AI Workflow

The complete virtual try-on workflow is shown below:



↓
Final Try-On Output

This workflow matches the uploaded system overview.

4. Input Stage

Objective

Prepare input images before AI inference.

Inputs used:

User Image

Contains:

- Body structure
- Pose information
- Background
- Lighting conditions

Garment Image

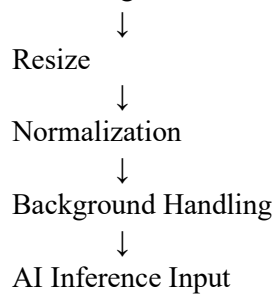
Contains:

- Clothing texture
- Pattern information
- Shape structure
- Category information

The workflow notes state user and garment images are processed before inference.

Input Flow

User Image + Garment Image



5. Preprocessing Module

Objective

Prepare images for downstream processing.

Operations performed:

- Image resizing
- Noise reduction
- Color normalization
- Background handling
- Contrast adjustment

Benefits:

- Better model accuracy
- Reduced noise
- Improved segmentation

6. Pose Detection Module

Objective

Detect body landmarks required for alignment.

Pose detection determines:

- Shoulder position
- Arm location
- Torso structure
- Body orientation

Technologies Used

OpenPose

OpenPose extracts multiple body keypoints.

Detected points:

- Shoulder
- Elbow
- Wrist
- Hip
- Knee

Advantages:

- High precision
- Better fitting accuracy

MediaPipe

MediaPipe provides lightweight pose estimation.

Benefits:

- Fast inference
- Real-time support
- Reduced latency

The workflow notes specify pose detection before parsing and warping.

Pose Flow

Input Image



Landmark Detection



Body Keypoints



Alignment Guidance

7. Human Parsing Module

Objective

Segment body regions for garment placement.

Human parsing separates:

- Hair
- Face
- Arms
- Torso
- Legs
- Background

The uploaded workflow notes mention separation of torso, arms, hair and background.

Technologies

Detectron2

Used for:

- Instance segmentation
- Body separation
- Mask generation

SCHP Parsing

Improves:

- Human segmentation
- Layering quality
- Occlusion handling

Parsing Flow

Input Image



Body Segmentation



Mask Generation



Region Extraction

8. Garment Warping Module

Objective

Align garments according to body posture.

Warping transforms garment structure according to:

- Pose
- Body size
- Shape
- Orientation

TPS Warping

Thin Plate Spline (TPS) performs deformation.

Capabilities:

- Sleeve fitting
- Body adaptation

- Flexible transformation

DensePose Mapping

DensePose improves:

- Surface understanding
- Alignment quality
- Body mapping

The uploaded workflow notes mention TPS warping for garment alignment.

Warping Flow

Garment Image



TPS Warping



DensePose Mapping



Aligned Garment

9. Rendering Module

Objective

Generate realistic try-on output.

Rendering improves:

- Texture quality
- Realism
- Visual consistency
- Color correction

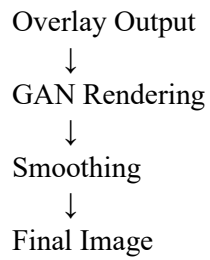
GAN Rendering

GAN-based rendering performs:

- Texture refinement
- Artifact reduction
- Overlay correction
- Realistic output generation

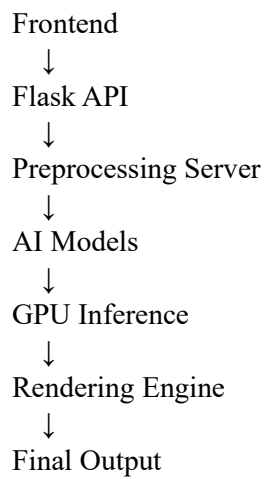
The uploaded workflow notes state GAN rendering improves realism.

Rendering Flow



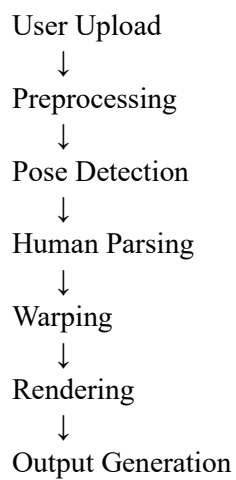
10. System Architecture

Proposed deployment structure:



The uploaded notes recommend frontend + Flask + GPU inference architecture.

11. Input / Output Flow Diagram



12. Challenges

Major workflow challenges:

Image Issues

- Lighting variation
- Background complexity
- Noise

Human Issues

- Pose diversity
- Occlusion
- Body variation

System Issues

- Latency
- GPU dependency
- Scalability

The uploaded workflow notes identify lighting, pose diversity and occlusion as major issues.

13. Future Improvements

Planned enhancements:

- Real-time optimization
- Better segmentation models
- Improved warping
- Lightweight inference
- 3D integration
- Better rendering models
- Multi-platform deployment

14. Conclusion

The AI workflow documentation defines the complete operational structure of the virtual try-on system from image upload to final rendering.

The workflow establishes a scalable architecture by integrating preprocessing, pose detection, human parsing, garment warping, and rendering modules, providing a strong foundation for future optimization and deployment.